

MATTHEW EDGAR

+1 (717) 668-5610 – matthew.c.edgar@icloud.com – [LinkedIn](#) – matthewcedgar.com – U.S. Citizen

EDUCATION

University of California, Berkeley

Bachelor of Science in Chemical Biology – Concentration in Computational Chemistry

August 2024 – May 2026

San Diego Mesa College

Associate of Science in Chemistry

Associate of Arts in Math and Pre-Engineering Math

Graduated with Honors

January 2020 – May 2024

Relevant Coursework:

- **Programming:** C/C++ Programming, Python Programming for Engineers, Computational Chemistry, Numerical Analysis (MATLAB)
- **Mathematics:** Discrete Mathematics, Mathematical Tools for the Physical Sciences, Linear Algebra, Differential Equations

EXPERIENCE

Williams Research Lab – University of California, Berkeley

Undergraduate Researcher

Berkeley, CA

January 2025 – May 2026

- Designed and assembled a nanoelectrospray apparatus, including custom circuitry and resistive components.
- Conducted 200 nanoelectrospray ionization experiments and developed Python scripts to collect and analyze their data.
- First author on a pending manuscript based on this research.

LIFEWest Ambulance Company

Emergency Medical Technician

Oakland, CA

May 2024 – June 2025

- Transported injured and critically ill patients to and from medical facilities across the San Francisco Bay Area.
- Logged over 500 hours in time-sensitive clinical environments.

T-Mobile

Mobile Expert

San Diego, CA

October 2021 – May 2022

- Assisted customers with mobile devices, wireless plans, and troubleshooting while providing individualized connection plans.
- Processed sales, device upgrades, activations, and account changes while meeting customer-service and sales objectives.

United States Navy

Machinist's Mate (E-5)

San Diego, CA

June 2017 – June 2021

- Active-duty enlisted sailor in the engineering departments of two ships; the USS Ardent and the USS Makin Island.
- Operated and maintained shipboard engineering systems, including refrigeration, diesel engines, and reverse-osmosis systems.
- Maintained fuel and oil systems, weapons systems, and kept proper watchstanding logs for quality assurance.

PROJECTS

Nanoelectrospray Research

Williams Research Lab – University of California, Berkeley – Python

- Developed a Python script that extracts the RGB values of specific pixels in an MP4 and records them in a CSV.
- A separate script extracts the RGB values from the CSV, averages, and plots them as a function of time.
- By comparing these plots to RGB values at known acidities, pH values can be estimated at different time points and the buffering capacities of different chemical agents can be deduced. This is useful for applications in mass spectrometry.

Monte Carlo Sampler using Egg Box Potential

Python

- Created a Python script that uses the Monte Carlo method to estimate the egg box energy integral of a particle.
- This project further applies different forces to the particle and simulates the positions of two interacting particles.

Personal Website

HTML, Jekyll, CSS, GitHub Pages

- Developing a personal portfolio website to display my research, professional experience, and software projects.
- Built and deployed this site using GitHub Pages and Jekyll.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, MATLAB

Scientific Computing: NumPy, Matplotlib, Monte Carlo Methods, Numerical Estimation Methods, Scientific Data Analysis

Tools: Git, GitHub, VS Code, Jupyter Notebook